

MOD 303 Course Outline

MOD 303 — Introduction to CGE Modelling

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	40 (indicative band 36–45)
Shape	4 parts · 12 sessions · 7 laboratories · 4 assignments · 1 examination · 1 final project
Model and data	CDCGE - AICTEM Country-Disaggregated CGE (ENVISAGE 10.01 structure) · GTAP 11 (v11b), reference year 2017, values in USD million
Delivery	Instructor-led — online, blended or intensive in person
Prerequisites	MOD 301. MOD 302 is recommended but not required. Calculus to partial derivatives, and matrix algebra to inversion. No prior CGE experience is assumed.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

A computable general equilibrium model is a social accounting matrix with behaviour attached. This course attaches it: the nesting structure, the demand system, the trade aggregation, the tax instruments, the factor markets and the closure. It then calibrates the model on the SAM of MOD 301, replicates the benchmark, runs the one counterfactual in this extraction that solves, and spends as much time on the four that do not.

The model is calibrated on the Utopia SAM of MOD 301 and replicates its benchmark. Of the five counterfactuals in the extraction one solves, and the course reports that one and spends a whole part of the manual on the four that do not — because that ratio is what applied work looks like and because the failures turn out to be the more informative half. Cutting the convergence grid by region shows the focus economy solving none of its carbon runs while three peers solve almost all of theirs: the instrument has no base here, which is a substantive answer rather than a failure. All 51 registered findings are carried in the appendix.

The rule the course is built on

Read the termination status before you read a variable. A solver returns numbers whether or not it converged, and a variable lifted from a run that did not solve looks exactly like

one that did. Closure is the second half of the same discipline: it is a modelling choice that is not in the data, and in this course's only solved scenario it changes the sign of the result.

Who it is for

Economists and modellers who will run or commission a country model, and analysts who must read somebody else's CGE result and say what it establishes. It is the entry point to the CGE sequence and assumes no prior experience of one.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	What a model of this kind is	01–02	—	4.0
P2	The blocks: production, households, trade, government	03–06	—	8.0
P3	Factor markets, closure, solution and calibration	07–09	01–03	10.5
P4	Scenarios, results, and the runs that do not solve	10–12	04–07	12.5
	Assignment and assessment time			5.0
	Total	12	7	40.0

The 12 sessions and 7 laboratories, in teaching order, and the subject matter of each:

P1 What a model of this kind is — Sessions 01–02 · 4.0 h

- **01 What a CGE Model Is** — The benchmark as an equilibrium, not a dataset; What behaviour adds: substitution, and prices that move; What it cannot add: anything not in the SAM; Comparative statics, not forecasting; The three things every CGE result is conditional on; Where this model sits: CDCGE on ENVISAGE structure
- **02 The System of Equations** — From accounting identity to behavioural equation; Zero profit, market clearing, income balance; Prices and quantities as paired unknowns; Why the system is square; The numeraire, and why one price cannot move; Walras's law and the redundant equation

P2 The blocks: production, households, trade, government — Sessions 03–06 · 8.0 h

- **03 Production** — The CES function and its two parameters; The nesting structure, and why nesting is an assumption; Calibrating shares from benchmark cost shares; The scale constant, and

units; Leontief at the top: intermediates in fixed proportions; The factor shares of the reference economy

- **04 Households** — Cobb-Douglas, CES and the linear expenditure system; Subsistence quantities and supernumerary income; The Frisch parameter: what it is and where it comes from; Income elasticities and why they matter for incidence; The savings decision, and what the closure does with it; One representative household, and what that forbids
- **05 Trade** — Why perfect substitutes fail: the specialisation problem; The Armington aggregate and its elasticity; Constant elasticity of transformation on the supply side; The small-country assumption and world prices; The exchange rate as a relative price; Import penetration and export intensity in this economy
- **06 Government** — Which instruments exist, and which have a base; Effective rates: revenue over base; The rates in this model, sector by sector; Government consumption, and how it is closed; The deficit as a residual or as a target; What this database's fiscal coverage forbids

P3 Factor markets, closure, solution and calibration — Sessions 07–09 · 10.5 h · Laboratory 01–03

- **07 Factor Markets and Closure** — Full employment and fixed real wage; Mobile and sector-specific factors; Savings-driven and investment-driven closure; Fixed current account or fixed exchange rate; How a closure can reverse a result; Reporting the closure as part of the result
- **08 Solving the Model** — Complementarity: why not just solve equations; The square condition, checkable before any solve; PATH, and what its termination statuses mean; Warm and cold starts; Reading a solve log; The convergence grid for this model
- **09 Calibration and Replication** — The calibration sequence; What is inferred and what is assumed; The zero-iteration solve; Benchmark replication and its residual; What a failed replication says about the SAM; The parameter register for this model
- *Lab 01 Calibrate and Replicate* — Does a model calibrated on this SAM reproduce it?
- *Lab 02 Read the Calibrated Object* — What did calibration infer, and what did it assume?
- *Lab 03 The Elasticity Register* — How much of this model is data and how much is assumption?

P4 Scenarios, results, and the runs that do not solve — Sessions 10–12 · 12.5 h · Laboratory 04–07

- **10 Designing a Scenario** — Does the instrument have a base?; Shock size, and linearity that is not there; Single shocks and combinations; What to hold fixed while the shock runs; The counterfactual as a comparison, not a forecast; The five scenarios in this extraction
- **11 Reading a Result** — Macro aggregates first, then prices, then quantities; Real and nominal, and the deflator that separates them; Decomposing a sectoral change; The result that surprises: investment under a tax rise; Sensitivity to the assumed elasticities; Reporting a conditional result honestly
- **12 When It Does Not Solve** — Termination statuses and what each means; Non-convergence against infeasibility; Diagnosing a stalled solve; Warm starting, and why it is not a universal fix; The instrument with no base; What may be reported from a run that did not solve
- *Lab 04 Run the One Scenario That Solves* — What happens when direct income tax rises, and why?
- *Lab 05 Read a Non-Convergent Run* — Four scenarios did not solve. What may be said about them?
- *Lab 06 The Convergence Grid* — Where does this model solve, and where does it not?
- *Lab 07 Decompose a Result* — Which mechanism produced the sectoral pattern?

What a graduate can do

1. State what a CGE model adds to a SAM, and what it cannot add.
2. Write the production nest for a sector and say what each elasticity governs.
3. Calibrate share and scale parameters from a benchmark, and explain why the elasticities cannot be calibrated with them.
4. Set out a demand system, identify its assumed parameters, and say what the Frisch parameter does.
5. Explain the Armington and transformation structures and why a small open economy needs them.
6. Identify every tax instrument in a model, its base and its effective rate.
7. State a closure, say what it asserts, and show that a different closure can reverse a result.
8. Verify that a calibrated system is square and replicates its benchmark, and interpret the residual.
9. Design a counterfactual, run it, and read the termination status before any variable.
10. Decompose a solved result into the mechanisms that produced it.
11. Diagnose a non-convergent run and say what, if anything, may be reported from it.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 120 items, used for formative practice and drawn on by the theory paper, which carries the marks
Laboratory exercises	35 %	7 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	The calibration report; The closure experiment; The non-convergence diagnosis; The model note
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions
Final project	25 %	A complete CGE study on this model: a policy question, a specified counterfactual, a stated closure, the run, the result or the diagnosis of its failure, and a nine-section model note

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No restricted data and no household record. The course carries one representative household by construction, so it produces no distributional result of any kind — that is school gap M-05 and it is the reason MOD 309 exists. Every elasticity in the model is an ENVISAGE default rather than an estimate for this economy, and no figure here should be cited as a parameter for Utopia.

Materials and set-up

A 155-page Technical Manual, 13 lecture decks of 133 slides, 7 laboratory decks of 86 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 120-item quiz bank, 5 assessment files, 15 datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

Learners need CDCGE instantiated at D:/EDIP/models/CDCGE, the PATH solver licence in place, and Julia started with `-pkgimages=no`. The pre-flight in the Instructor Guide has six checks and all six have failed on a teaching machine at least once; run it before the first laboratory rather than during it.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in `COURSE_VERSION.json`.

MOD 303 Introduction to CGE Modelling · version 1.0 · AICTEM EDIS Academy, School 03 Economic Modelling.